Notice, question, predict

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use observations to make a prediction that can be tested.

SUCCESS CRITERIA - I can...

1. Separate stated evidence from unsupported claims.
2. Extend a position pattern and state an assumption.
3. Name an observation that could test the prediction.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Read a scale

A ruler shows a toy car at 2 cm, then at 7 cm. Draw both positions and the change. How far apart are the marks?

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Delta x = x_f - x_i$; $x_{next} = x_{now} + \Delta x$.
For a repeating position pattern only. i = initial; f = final. Check the pattern against evidence.

Worked reasoning: A cart is at 2, 5 and 8 m at 0, 1 and 2 s. Equal jumps of 3 m suggest 11 m at 3 s, if the pattern continues.

Your turn: Draw a position strip for these four times. Mark which point is a prediction rather than an observation.

Marzano check - circle: ① ② ③

03 Extract the evidence

Fictional briefing: “Repeated images show an object approaching Earth’s neighborhood. Its path and size remain uncertain.” Give two supported observations and one unsupported conclusion.

Marzano check - circle: ① ② ③

04 Fit a simple model

Use the supplied cart data. State a rule for position as a function of time and predict x at 5 s. Explain why this is a model, not a guarantee.

t (s)	x (m)
0	1
1	4
2	7
3	10

Marzano check - circle: ① ② ③

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Continue down each column.

05 Challenge the prediction

For the cart in 4, a new measurement at 5 s is 14.0 ± 0.2 m. Is it consistent with your exact constant-velocity prediction? Give two possible explanations.

Marzano check - circle: 1 2 3

06 Design the observation

You can request two new observations of the fictional asteroid. Specify what, when, and how each would improve a position prediction.

Marzano check - circle: 1 2 3

07 Compare claims

Claim A: “A straight-line model predicts where it may be.” Claim B: “The asteroid will hit Earth.” Explain what additional evidence and physics separate these claims.

Marzano check - circle: 1 2 3

08 Choose useful evidence

Two groups time the same toy car over the same marked track. One records three times; the other writes "pretty fast." Which record would let someone check the claim? Explain.

Marzano check - circle: 1 2 3

09 Exit check

Supplied positions are 12, 8, 4 m at 0, 2, 4 s. Predict x at 7 s. State the rate, your assumption, and a future observation that would test it.

Marzano check - circle: 1 2 3

10 STRETCH: Apply it another way

Return to the worked example in problem 2. Construct two position rules that match all three observations in the worked case but disagree at 4 s. Explain why the data do not guarantee either extrapolation.

Marzano check - circle: 1 2 3